

## EXPERIMENT 4

### Aim

To implement the K-Nearest Neighbors (KNN) algorithm on a real-world dataset and evaluate its performance using classification metrics.

### Problem Statement

Heart disease prediction is an important medical problem where patients must be classified as:

- Presence of heart disease (1)
- Absence of heart disease (0)

The objective is to:

- Build a KNN model to classify patients based on medical attributes
- Analyze the effect of the value of K on model performance
- Evaluate the model using appropriate performance metrics

### Dataset Link

- Dataset Name: Heart Disease Dataset
- Source: Kaggle
- Link: <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

### Methodology

#### *Data Collection*

The dataset (heart.csv) was loaded into the environment using Pandas.

It contains patient health parameters such as age, cholesterol, blood pressure, etc., along with a target variable indicating the presence or absence of heart disease.

#### *Data Preprocessing*

The dataset was checked for missing or inconsistent values to ensure data quality.

Features (X) and target variable (y) were separated, where:

- X contains independent variables (input features)
- y contains the dependent variable (output/label)

This step ensures the data is properly structured for model training.

### *Data Splitting*

The dataset was divided into:

- Training set (80%) → used to train the model
- Testing set (20%) → used to evaluate performance

Stratified sampling was applied to maintain the same proportion of classes in both sets, avoiding bias.

### *Feature Scaling*

Feature scaling was performed using StandardScaler, which transforms data to have:

- Mean = 0
- Standard deviation = 1

This is important for KNN because it is a distance-based algorithm, and unscaled features can dominate the distance calculation.

### *KNN Model Training*

The K-Nearest Neighbors (KNN) algorithm was used for classification.

- Value of K = 5 was chosen
- The model stores training data and classifies based on the majority class of nearest neighbors

Training was performed using the scaled training dataset.

### *Prediction*

The trained KNN model was used to predict class labels for the test dataset.

Predictions are based on the majority vote of the 5 nearest neighbors.

### *Model Evaluation*

The model performance was evaluated using multiple metrics:

- Accuracy Score → Overall correctness of predictions
- Confusion Matrix → Shows:
  - True Positives (TP)
  - True Negatives (TN)
  - False Positives (FP)
  - False Negatives (FN)
- Precision → Correct positive predictions out of total predicted positives

- Recall → Correct positive predictions out of actual positives
- F1-Score → Harmonic mean of precision and recall

These metrics provide a complete understanding of model performance beyond just accuracy.

### Output:

```
----- KNN Results (k = 5) -----
Accuracy: 0.8634146341463415

Confusion Matrix:
[[87 13]
 [15 90]]

Classification Report:
              precision    recall  f1-score   support

     0       0.85         0.87         0.86         100
     1       0.87         0.86         0.87         105

 accuracy          0.86         0.86         0.86         205
  macro avg       0.86         0.86         0.86         205
 weighted avg     0.86         0.86         0.86         205
```

### Conclusion:

In this experiment, the K-Nearest Neighbors algorithm was implemented on the Heart Disease dataset. Feature scaling was applied to improve distance calculations, and the model was evaluated using accuracy and classification metrics. The results demonstrate that KNN is an effective classification technique for medical datasets when an appropriate value of K is chosen.